Summary

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A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$, for all x in I.

If F is an antiderivative of f on an interval I, then the function f has a whole **family of antiderivatives**.

If a function G is an antiderivative of f on I, then $G(x) = F(x) + C$, for all x in I, for some constant C.

The family of of *all* antiderivatives of f is denoted by $\int f(x)\,dx$ and called **indefinite integral of** f. Therefore, if F is an antiderivative of f, $\int f(x)\,dx = F(x) + C$.

Basic Indefinite Integrals

- $\int k\,dx = kx + C$
- $\int \frac{1}{x}\,dx = \ln|x| + C$
- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \qquad (n \neq -1)$
- $\int e^x\,dx = e^x + C$
- $\int a^x\,dx = \frac{a^x}{\ln(a)} + C$
- $\int \cos(x)\,dx = \sin(x) + C$
- $\int \sin(x)\,dx = -\cos(x) + C$
- $\int \sec^2(x)\,dx = \tan(x) + C$
- $\int \csc^2(x)\,dx = -\cot(x) + C$
- $\int \sec(x)\tan(x)\,dx = \sec(x) + C$
- $\int \csc(x)\cot(x)\,dx = -\csc(x) + C$
- $\int \frac{1}{x^2+1}\,dx = \arctan(x) + C$
- $\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin(x) + C$